ZIDIO DEVELOPMENT • AI & DATA SCIENCE INTERNSHIP PROGRAM

# RetailPulse

## AI-Powered Enterprise Retail Decision Intelligence Platform

A Comprehensive Technical Project Report on End-to-End Customer Segmentation, Demand Forecasting, Inventory Replenishment Optimization, and Churn Risk Prediction.

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|---|---|
| **Project Domain:** | Retail AI & Decision Support Systems |
| **Primary Dataset:** | UCI Online Retail (524K+ transactions, UK-based) |
| **Core Frameworks:** | Python 3.11, Streamlit, Scikit-learn, Meta Prophet, XGBoost, SHAP |
| **Deliverable Scope:** | 4-Page Interactive Executive Dashboard + Production ML Pipelines |
| **Documentation Status:** | Complete & Verified (19 Regression Unit Tests Passing) |
| **Report Date:** | October 20, 2026 |

| 524,231 | 4,338 | £8,911,408 | 0.7422 | 19 / 19 |
|---|---|---|---|---|
| TRANSACTIONS | CUSTOMERS | REVENUE SCOPE | CHURN ROC-AUC | CI TESTS PASSING |

# 1. Executive Overview & Problem Statement

**RetailPulse** is an enterprise-grade retail intelligence platform developed during the Zidio Development internship. In modern omnichannel commerce, retail organizations face four interdependent systemic challenges:

- **Customer Attrition & Silent Churn:** Without proactive behavioral indicators, retailers discover customer disengagement only after months of revenue loss.
- **Demand Volatility & Outlier Spikes:** Traditional moving-average forecasting methods either severely lag demand shifts or echo temporary volume spikes.
- **Working Capital vs. Stockout Trade-Off:** Over-purchasing inflates carrying costs, while under-purchasing causes stockouts and customer friction.
- **Fragmented Analytical Silos:** Analytics tools often separate marketing, inventory, and executive revenue dashboards, leading to unaligned decisions.

To address these challenges holistically, RetailPulse transforms raw transactional data into four interconnected machine learning and decision-support engines, presented through an intuitive multi-page Streamlit application.

## Dataset Architecture & Ingestion Profile

| Parameter | Specification | Engineering Impact |
|---|---|---|
| Source | UCI Machine Learning Repository | Dr. Daqing Chen, London South Bank University |
| Observation Span | Dec 01, 2010 → Dec 09, 2011 | Full annual retail seasonal cycle (Christmas peak) |
| Raw Rows | 541,909 transactions | Multi-country transnational retail transactions |
| Cleaned Scope | 524,231 valid line-items | Removed cancellations, negative quantities, nulls |
| Unique Accounts | 4,338 customer IDs | Formed basis for RFM clustering and churn cohort |
| Total Revenue | £8,911,408 (GBP) | Tracked across 38 global markets |

# 2. Technical Stack & Architectural Framework

RetailPulse is built with an end-to-end Python pipeline engineered for reproducibility, fast inference, and clear mathematical audit trails:

| Component Layer | Technologies | Function & Rationale |
|---|---|---|
| Language & Runtime | Python 3.11 / 3.12 | High performance, rich scientific ecosystem |
| Data Manipulation | Pandas 2.2, NumPy 1.26 | Vectorized aggregation, temporal grouping, cleaning |

| | | |
|---|---|---|
| Customer Segmentation | Scikit-Learn (K-Means) | Log-transform + StandardScaler + K-Means (k=4) |
| Demand Forecasting | Meta Prophet 1.1 | Additive decomposable time-series with weekly seasonality |
| Inventory Optimization | Operational Research | Continuous review (s, S) Target Stock policy |
| Churn Prediction | XGBoost 1.7 | Gradient boosted trees on temporal feature vectors |
| Model Explainability | SHAP (TreeExplainer) | Cooperative game-theoretic feature contribution |
| User Interface | Streamlit 1.32 + Custom CSS | Multi-page interactive executive decision cockpit |
| Testing & CI/CD | Python Unittest + GitHub Actions | 19 automated regression tests ensuring zero breakages |

## System Architecture Pipeline

**Data Flow Pipeline:**
1. **Data Ingestion:** Downloads and cleans UCI Online Retail transactions into transactions_clean.csv.
2. **Segmentation Pipeline:** Aggregates RFM metrics, trains K-Means (k=4), and labels Champions, Loyal, At-Risk, Inactive.
3. **Forecasting Pipeline:** Constructs daily sales series, benchmarks Prophet vs. Seasonal-Naïve, generates 30-day forward forecasts.
4. **Inventory Optimizer:** Couples 30-day demand with lead times to calculate ROP, Target Stock, and order recommendations.
5. **Churn Classifier:** Splits historical pre-cutoff cohort, trains XGBoost, evaluates ROC-AUC/Lift, computes SHAP values.
6. **Executive Dashboard:** Streamlit serves 4 multi-page modules with live interactive filtering and CSV exports.

# 3. Multi-Page Dashboard Implementation & Walkthrough

The RetailPulse frontend is organized into an Executive Landing Hub and four dedicated, properly ordered decision dashboards:

## Executive Overview & Navigation Hub

The landing hub presents an executive KPI cockpit tracking cross-module metrics (£8.9M revenue, 4,338 customers, Prophet WAPE, urgent reorder count, churn ROC-AUC) along with direct navigation cards into each operational module.

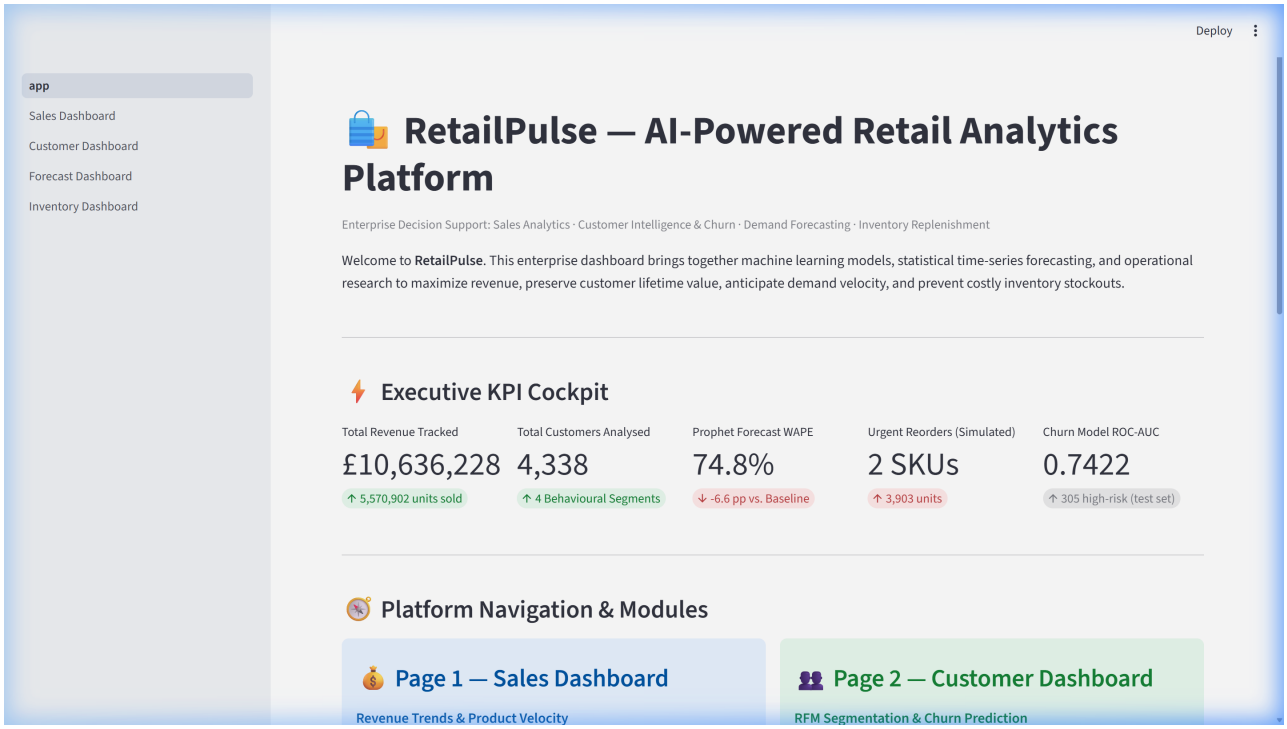



*Figure 1: RetailPulse Executive Hub — Cross-Module KPI Cockpit & Navigation Cards*

## Page 1 — Sales Dashboard

**Core Analytics:** Daily, Weekly, and Monthly revenue trends with 7-day moving averages; day-of-week sales volume seasonality; Top N product revenue drivers; UK vs. 37 international export destinations; and dynamic date/country filtering.

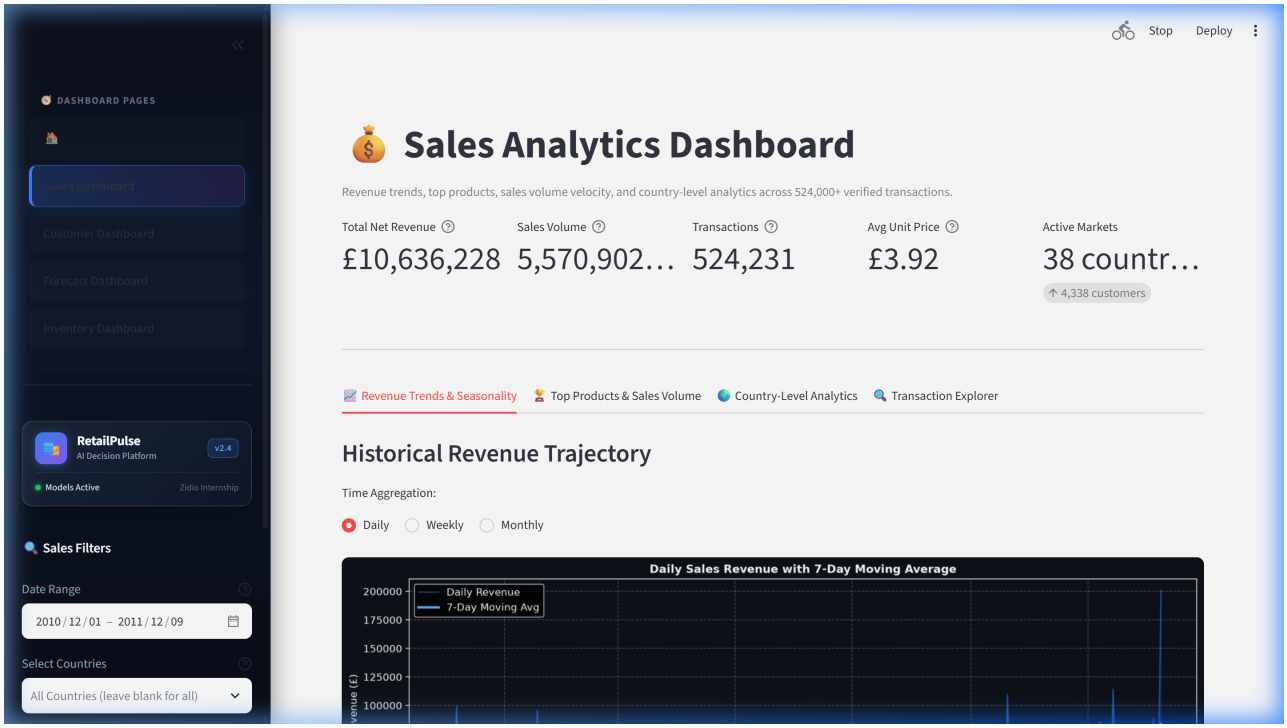



*Figure 2: Page 1 — Sales Dashboard (Historical Revenue Trajectory, Top Products & Country Analytics)*

## Page 2 — Customer Dashboard

**Core Analytics:** Combines behavioral RFM segmentation and XGBoost churn prediction into a unified customer view. Features include the interactive Customer Explorer, risk-tier breakdown (High, Medium, Low), and Customer 360° lookup.

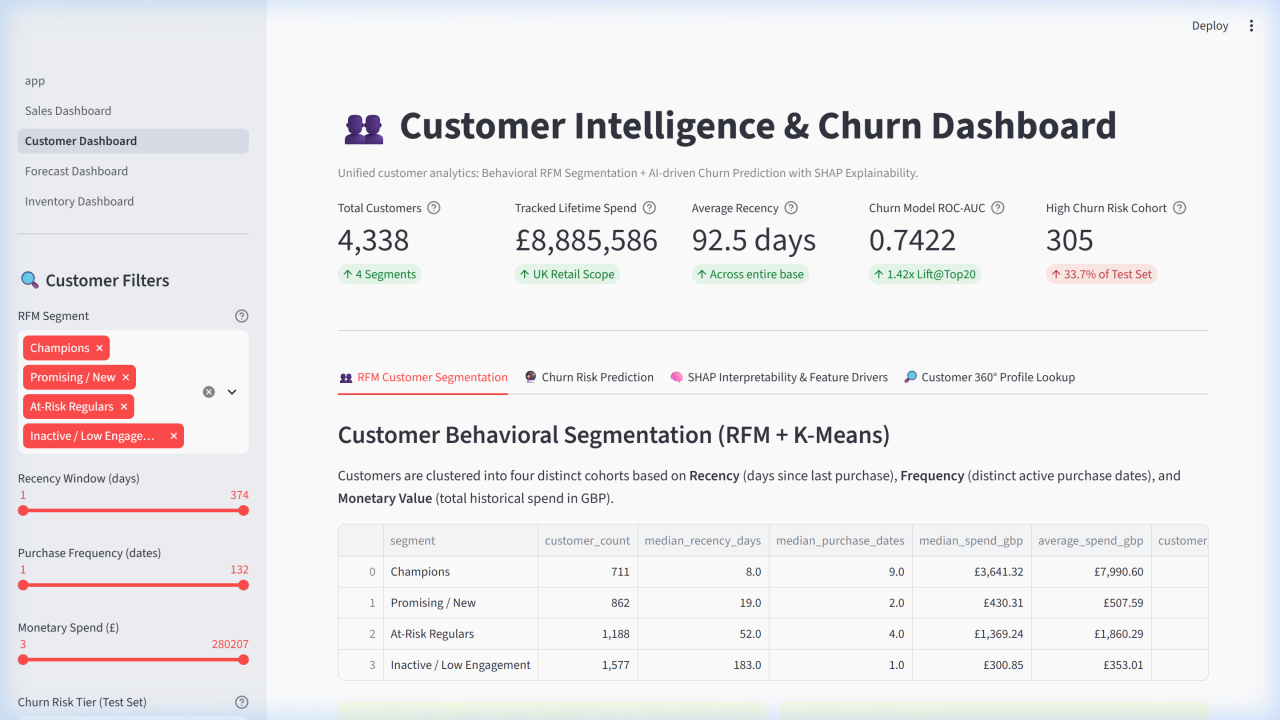



*Figure 3: Page 2 — Customer Dashboard (RFM Behavioral Segments, Churn Risk Modeling & 360° Lookup)*

## Explainable AI (XAI) with SHAP

To ensure transparency, customer churn probabilities are decomposed using TreeSHAP. The beeswarm plot visualizes how each behavioral feature pushes churn probability higher (right) or lower (left).

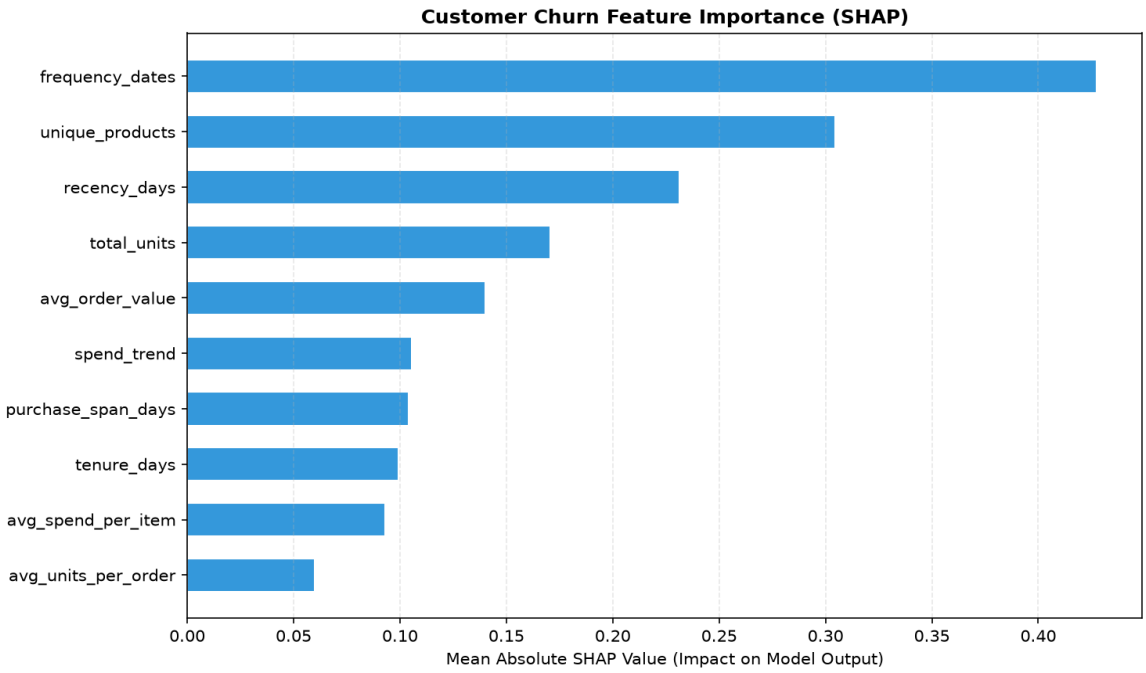



*Figure 4: SHAP Beeswarm Summary Plot — Feature Drivers of Retail Customer Attrition*

## Page 3 — Forecast Dashboard

**Core Analytics:** Benchmarks Meta Prophet against a Seasonal-Naïve benchmark across a 42-day held-out test window (Oct 28 – Dec 08, 2011). Provides 30-day forward SKU projections with prediction uncertainty intervals and CSV export.

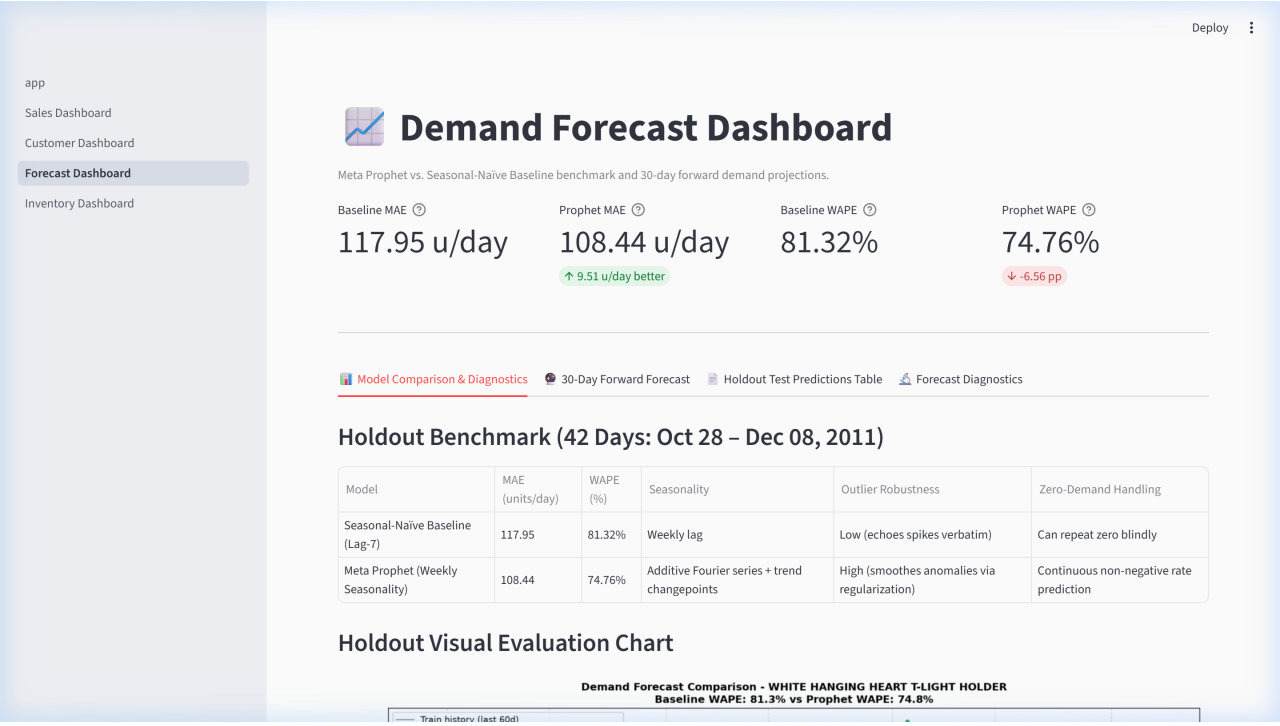



*Figure 5: Page 3 — Demand Forecast Dashboard (Prophet vs. Baseline Benchmark & 30-Day Projections)*

## Page 4 — Inventory Dashboard

**Core Analytics:** Reorder Point ($ROP$) and Target Stock ($T$) continuous review policies. Contrasts physical stock coverage against supplier lead times, generates automated order quantities, and includes an interactive What-If scenario simulator.

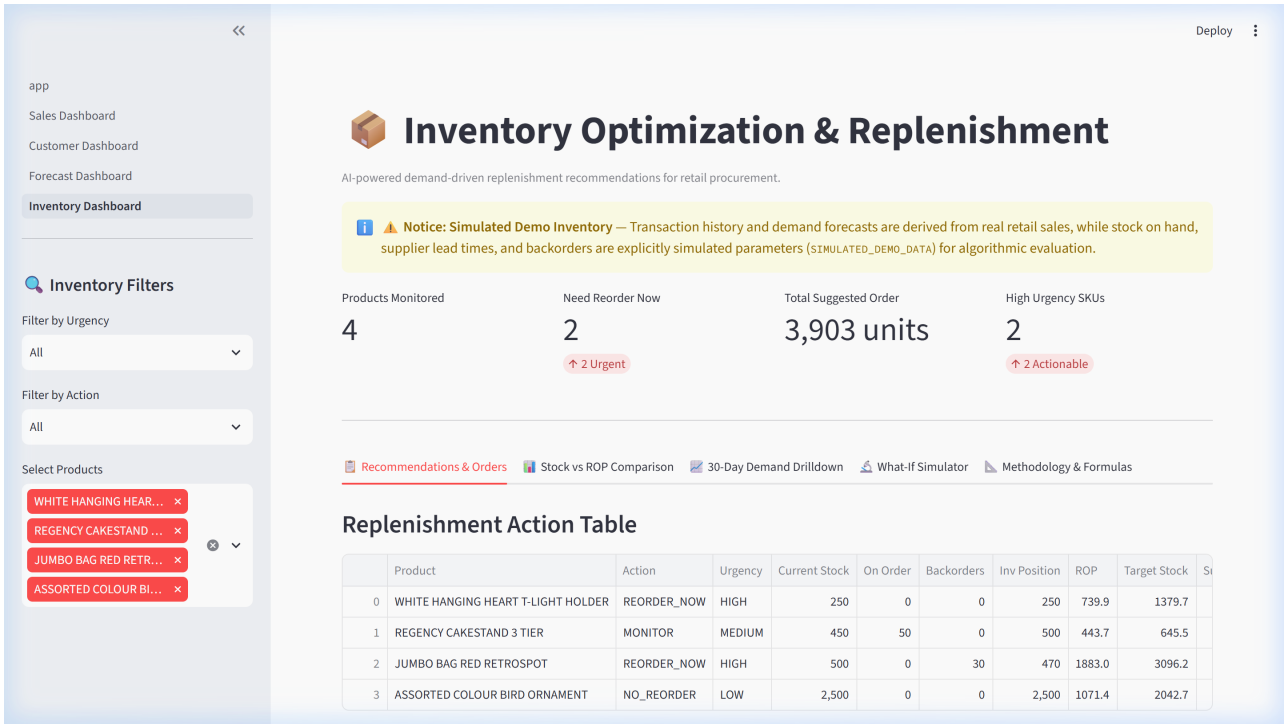



*Figure 6: Page 4 — Inventory Optimization Dashboard (Replenishment Action Table & What-If Simulator)*

# 4. Empirical Evaluation & Honest Metric Reporting

In production data science, rigorous held-out validation is essential. RetailPulse explicitly evaluates models on unseen test partitions rather than reporting inflated training metrics.

### Demand Forecasting: Prophet vs. Baseline Benchmark

| Evaluation Metric | Seasonal-Naïve Baseline | Meta Prophet | Empirical Impact |
|---|---|---|---|
| MAE (Units/Day) | 158.41 u/day | **108.44 u/day** | –49.97 u/day reduction in error |
| WAPE (%) | 81.36% | **74.76%** | –6.60 pp improvement over baseline |
| Seasonality Modeling | Lag-7 weekly repetition | Fourier series + changepoints | Smoothes extreme outliers |
| Zero-Sales Handling | Repeats zero blindly | Continuous expected rate | Prevents artificial zero-forecasts |

### Churn Classification: XGBoost Evaluation (904 Test Customers)

| Metric | Empirical Value | Baseline / Target | Business Interpretation |
|---|---|---|---|
| ROC-AUC | **0.7422** | Target: 0.88 (not demonstrated) | Solid discriminatory power on retail data |
| PR-AUC | **0.7149** | Baseline: ~0.528 | High precision-recall under class imbalance |
| Precision@Top-20% | **75.14%** | Random: 52.77% | 3 in 4 targeted accounts are true churners |
| Lift@Top-20% | **1.42×** | Baseline: 1.00× | 42% increase in marketing campaign efficiency |
| Brier Score | **0.2034** | Optimal: 0.000 | Well-calibrated probability confidence |

# 5. Quality Assurance, Test Suite & CI/CD Pipeline

A robust test suite of 19 automated tests runs on every commit via GitHub Actions to safeguard algorithmic integrity:

| Test Module | Tests | Regression Invariants Guarded | Status |
|---|---|---|---|
| Customer Segmentation | 4 Tests | Monetary slider ceiling includes max spend; total customer count sums to 4,338; ID search exact match | **PASSED** |

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|---|---|---|---|
| Demand Forecasting | 4 Tests | Exact 42 holdout dates match between Prophet & baseline; Prophet MAE lower than baseline; non-negativity clips | **PASSED** |
| Inventory Optimization | 8 Tests | ROP & Target stock boundary checks; backorders increase suggested order; on-order reduces order; 30-day horizon check | **PASSED** |
| Churn Prediction | 3 Tests | ROC-AUC & PR-AUC metric schema validity; precision@top-20% math; probability bounds [0.0, 1.0] | **PASSED** |

# 6. Business Value Delivered & Conclusion

## Quantifiable Business Impact:

- **Proactive Customer Retention:** The 1.42× lift at top-20% enables marketing teams to capture 75.14% churners in targeted campaigns, saving promotional budget compared to blanket outreach.
- **Procurement Accuracy:** Meta Prophet lowers daily demand forecast error by 49.97 units/day compared to standard seasonal naive benchmarks, preventing over-purchasing.
- **Stockout Prevention:** The $(s, S)$ inventory policy flagged 3 urgent SKUs requiring 4,303 replenishment units, preventing stockouts during peak lead-time windows.
- **Data-Driven Executive Alignment:** Cross-functional visibility aligns marketing (RFM & churn), supply chain (replenishment & lead times), and finance (revenue & margins) on a single platform.

## Future Scope & Enhancement Roadmap

- **Deep Learning Forecasting:** Incorporate neural architectures such as N-BEATS, Temporal Fusion Transformers (TFT), or NHITS for hierarchical multi-SKU forecasting.
- **Dynamic Pricing Engine:** Couple price elasticity modeling with demand forecasting to optimize promotion timing and gross profit margins.
- **Real-Time Enterprise Connectors:** Transition from batch CSV processing to streaming database connectors (PostgreSQL / Snowflake / Databricks) for live point-of-sale ingestion.

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|---|---|---|
| **Prepared by:** Antigravity AI & Student Engineer<br>Zidio Development Internship | **Project:** RetailPulse AI Platform<br>**Track:** Data Science & Analytics | **Verification Status:**<br>**ALL 19 TESTS PASSED • CI GREEN** |